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CMP6214 User Experience Design

Journey Finder (Prototype)

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1 Introduction

Being tasked to design a mobile application for “Tech for the People of West Midlands”, I settled on the idea of a tourist app, based in Birmingham, that is applicable to those who like to travel or even locals. Through this app, it will assist in growing Birmingham’s tourist, local businesses, and potentially the health of those who might be a user. The name of the app is “**Journey Finder**”.

Journey Finder will be different to those which are already out there, like Tripadvisor and Yelp, by somewhat combining these two ideas into one. Tripadvisor is a really good app that allows its users to leave reviews for other travellers and Yelp is a great will promoting businesses through reviews. So having one app that will help businesses grow through reviews, allow hotspot locations to be recommended to users and an in-app reward system that will benefit the users, I think it is an app that many tourists and locals would like to have.

2 User Experience Design

2.1 User Personas

Within this section, it will go through the three unique user personas, including one with some sort of impairment, who are targeted for this mobile application. Personas are a crucial tool when it comes to User Experience (UX) Design, it allows the designer to think as other people by creating a fictional character to understand the goals and behaviours of a person that may use their product (Dam and Siang, 2019).

1st Persona:

The first user persona is a 21 year old, university student, Makoto Niijima. She moved in from Japan to England to study Business Management. She is Extroverted, likes to save money and loves finding new places to visit. Always on the lookout for a healthier life style and meeting new people through her travels.

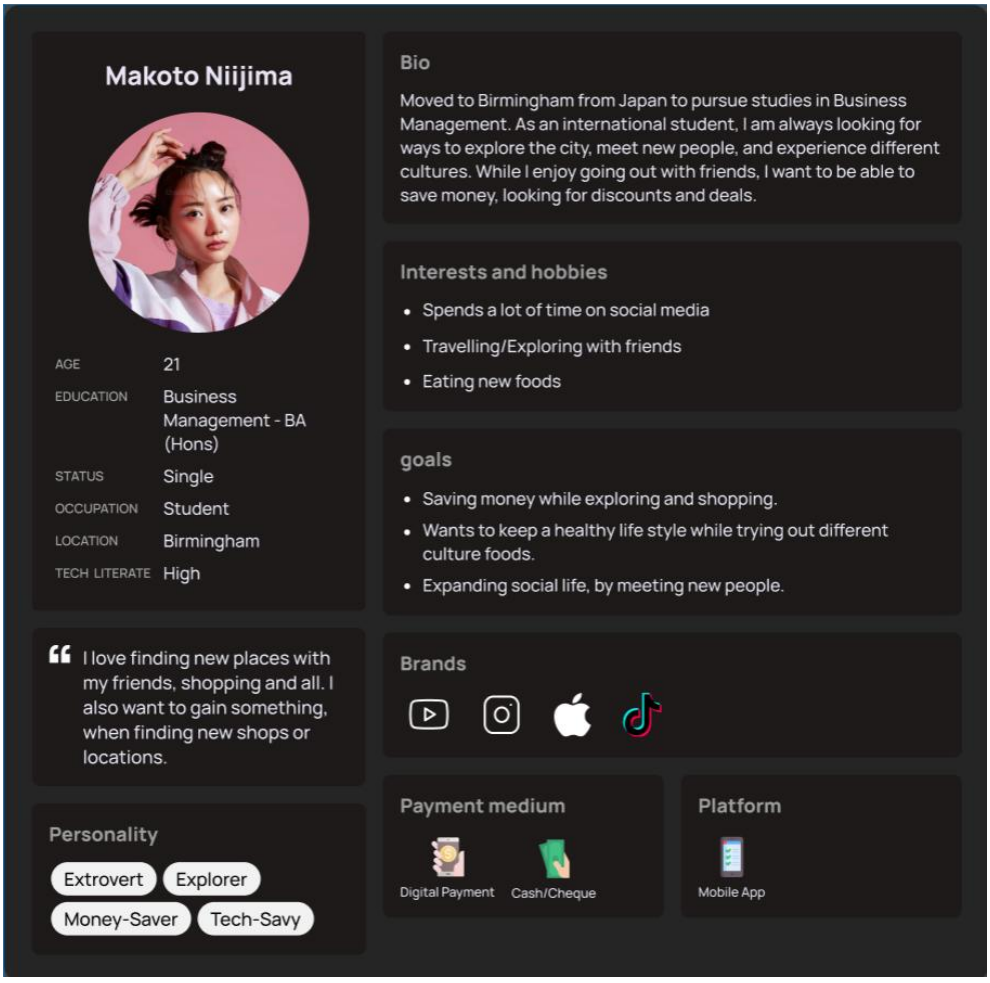


Figure 1 User Persona: Makoko Niijima

2nd Persona:
Richard Zato, a 45 year old, married photographer. He is a freelance photographer, who is always looking for places to visit, outside of his work, to capture and relax. He's always lived in the UK, capturing its landscapes and cultural diversity. Richard has always wanted to expand his portfolio and ways to increase and improve his camera equipment. He's looking for an app that will do both.

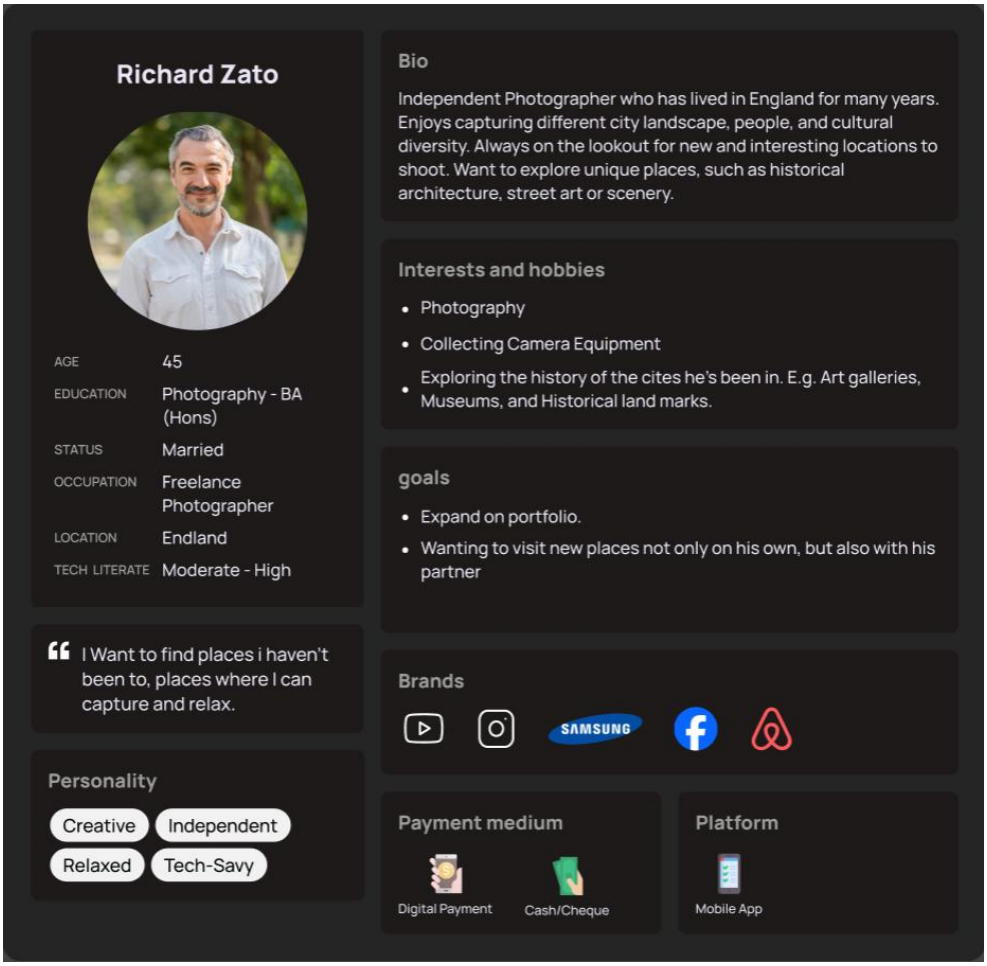


Figure 2 User Persona: Richard Zato

3rd Persona:
For the final persona, Carl Matthews is a 36 year old, married man. He is a new shop owner, in a popular area of Birmingham, running it with his family. He is a passionate and creative person, who takes part in the art of pottery. He is red and green colour blind (Deuteranopia). He wants to meet new people and get to know them through their culture. Carl is searching for a way to grow his business, to get more eyes on his shop, through the internet.

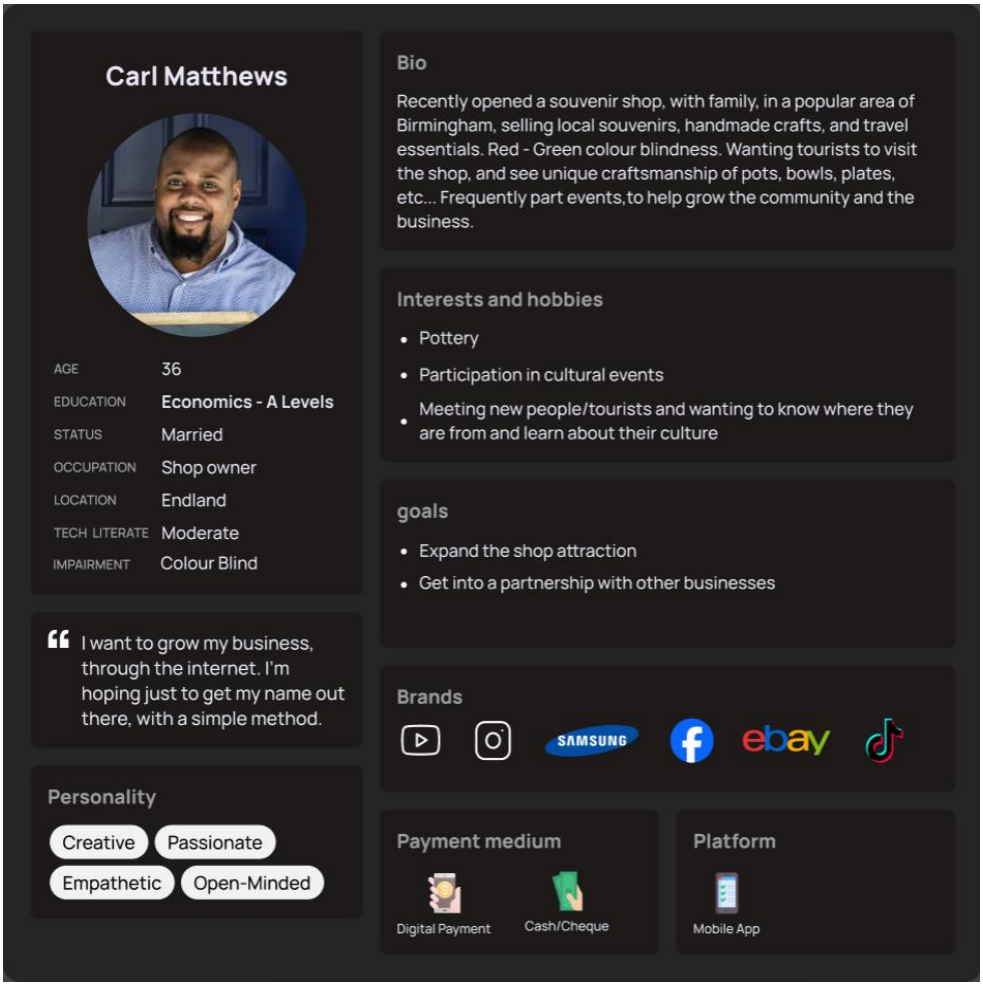


Figure 3 User Persona: Carl Matthews

2.2 User Scenarios

User scenarios are a way for the UX designer to make a situation for their personas, so it can give them an overall analysis of what are the requirements needed to make the application fit for different user behaviours (Interaction Design Foundation, 2019).

Makoto Niijima:
Makoto just finished up her final assignment before the winter holiday. She wants to treat herself, by celebrating with her friends and going to a place in Birmingham that she has never eaten at before. However, since she is a student, she is wary of her budget.
She came along an app, Journey Finder. She downloads the app and visits the home page, where finds the restaurants section and within this it shows that there is an offering of a 10% discount to those who have made an account. She quickly

makes an account. After the dinner with her friends, Journey Finder suggests a location that she has never been to, a place filled with Birmingham murals, so she excitingly visits this place. Here she comes across a photographer, which allows her to interact with someone new. The app then recommends a new shop in town, which she curiously visits.
After the day is over, she checks Journey Finder, since the app is connected to her phone location and she has made an account, the suggested locations that she visited, allowed her to accumulate points, that can be redeemed for gift cards.

Richard Zato:
Richard is planning to head out to Birmingham, a place he has rarely visited, to expand on his portfolio. Since he doesn't know much about the place, he searches online for areas that he can visit. This is where is finds Journey Finder.
Once he signed up, he discovered locations where he can take pictures of different landmarks and points of interest. He plans his route using the in-app map. He walks across canals, murals and events, which excites him and allows him to capture picturesque scenes. Using the community page, he can share his passion with others.
Since he visited these areas, he gained in-app points and enough to use them on a coupon, for a store that sells camera equipment. Thus, allowing him to fulfil his hobby of collecting.

Carl Matthews:
Carl just opened his souvenir shop, for tourists and curious buyers, in the midst of a popular area in Birmingham. However, he's not getting a lot of customers. To get his name out there, he looks for apps to help with this, he finds Journey Finder. Looking through the preview images, it's easy on his eyes (due to his colourblindness) and simple to use.
Once signed in, he checks his account page, where it shows a feature for a business account. After readding up what it does, he finds out that paying a small subscription fee, he can set up business plan, where the app will push his shop up the recommendations so more people can see it.
After a week of buying the subscription, he has realized that more people are visiting the shop and it's not just tourists. As some students walk in, his personality of wanting to know about them and their culture kicks in, leading to a nice conversation.

2.3 User Requirements

Using the MoSCoW method, this section will list out the needed requirements for a good user experience, requirements that are key but not needed, some that are there for slight user experience and things that the application won't have.

<u>Must Have:</u>

- User reward system
- In-app map
- Login system
- Location recommendations
- Offline mode (premium feature)
- Business account (premium feature)
- Location images

- Payment System

Should Have:

- Notifications on events
- Community system for sharing photos
- Filter to fine tune what you want
- Multi-Language support
- One-hand usage
- Responsive layout
- Itineraries (premium feature)
- More Rewards (Premium feature)
- History (previously visited)
- Personalised account page

Could Have:

- Private messaging system (not community system)
- Custom itineraries
- Augmented Reality (AR) feature

Won't Have:

- Booking system
- Detailed information

2.4 Hierarchical Task Analysis

Here will show an HTA flowchart that will describe what users will do to get to a certain goal through the various tasks. Not all will be in the high-fidelity.

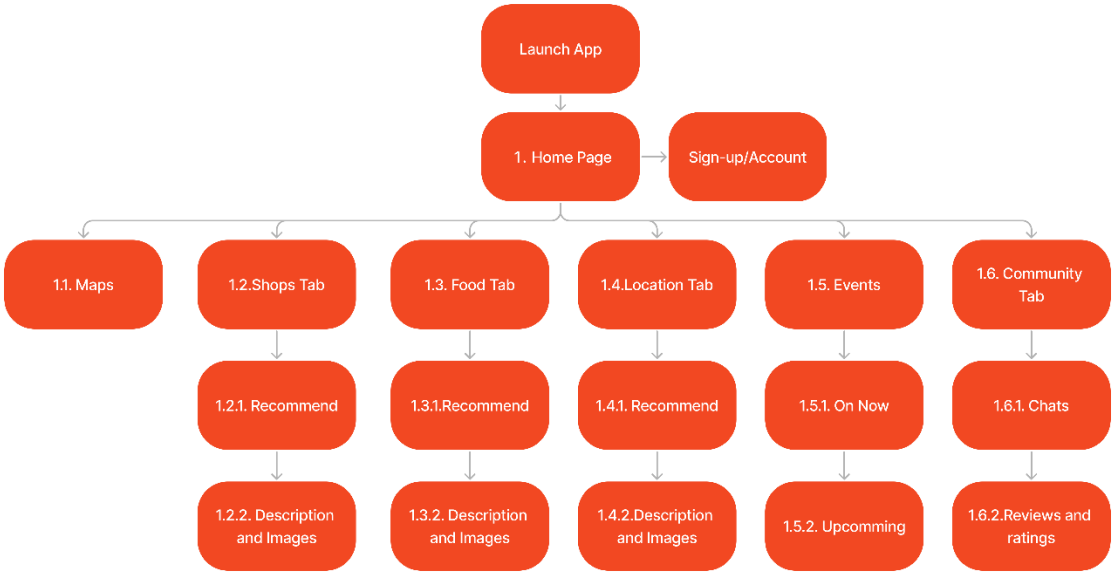


Figure 4 Home Page HTA

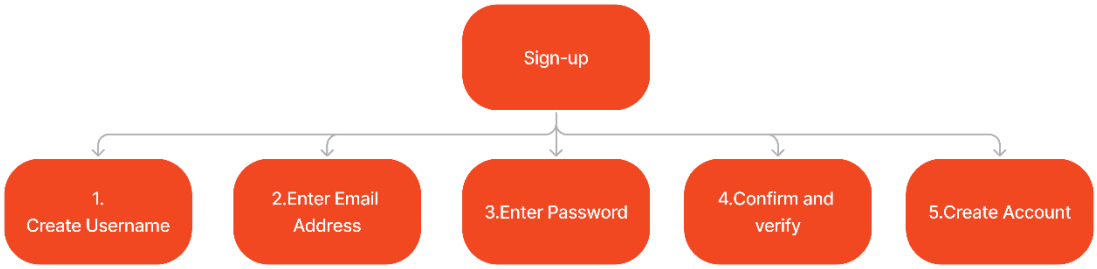


Figure 5 Sign Up Page HTA

Figure 6 Subscription HTA

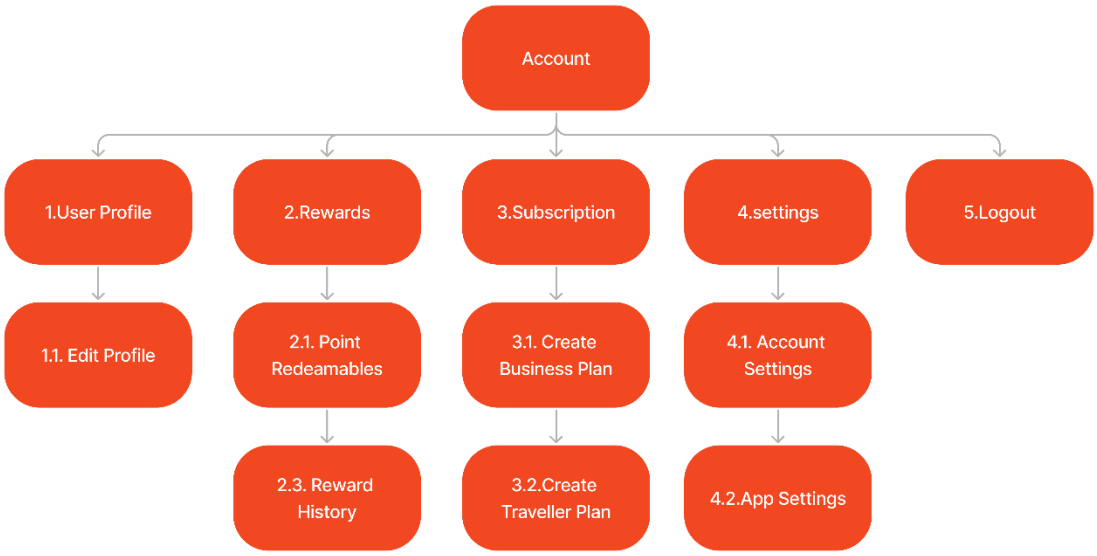


Figure 7 Profile Page HTA

2.5 Paper prototypes (low-fidelity)

Login Page:

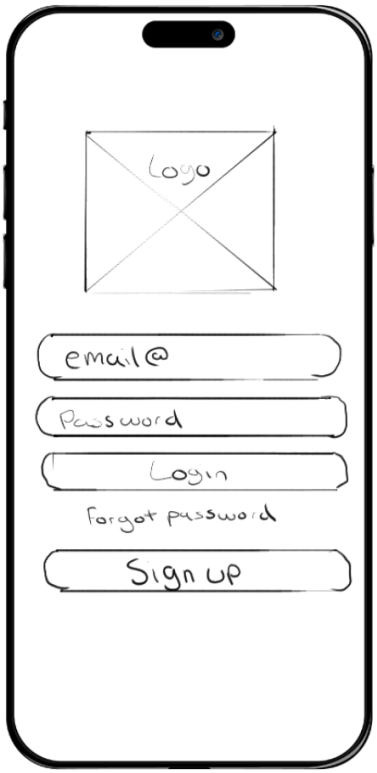


Figure 8 Low-Fidelity Login

All paper prototypes were done in Photoshop.

When sketching the low fidelity for the login page, my main thought was, what makes a login page. Email, Password and a login button, are what first came to mind. (Soegaard, 2024) has a great page for UX designers for login examples pages. Using this I had a great idea of what a good login page should be.

Following Hick's Law, having minimal buttons and choices within the login screen would be important for quick access to the app, due to the small number of options leading to less effort and time when deciding (Nikolov, 2017).

Home Page:

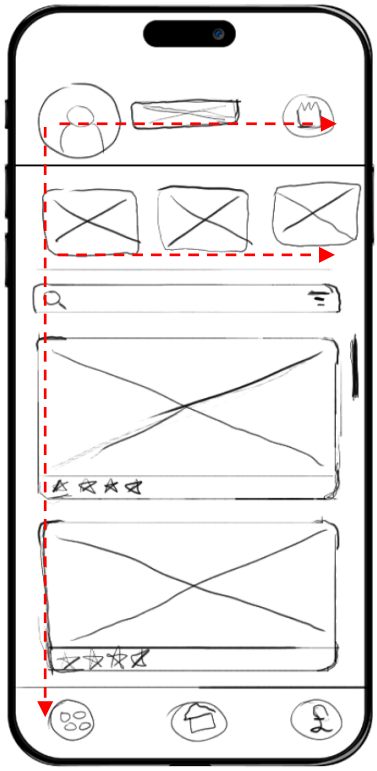


Figure 9 Low-Fidelity Home

At this stage of the design going through different websites and looking through what others are like, this is something I did very often. Here I just wanted a main hub for the user to have access to the recommended locations and events, and a means to access key pages to the app.

Visual Hierarchy, using this principle will allow me to have the user look at what is of importance first. Like the profile picture to the premium icon, then to the various tabs and continuing down to the navigation bar. The F pattern was what I tried to implement, since this page contains a lot of images (Soegaard, 2019).

The reason for having the navigation bar at the bottom was to allow the user to have easy access to the key pages of the app, with one hand.

Community Page:

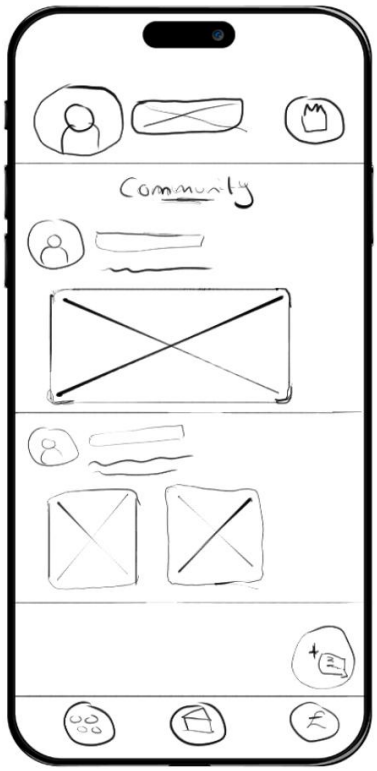
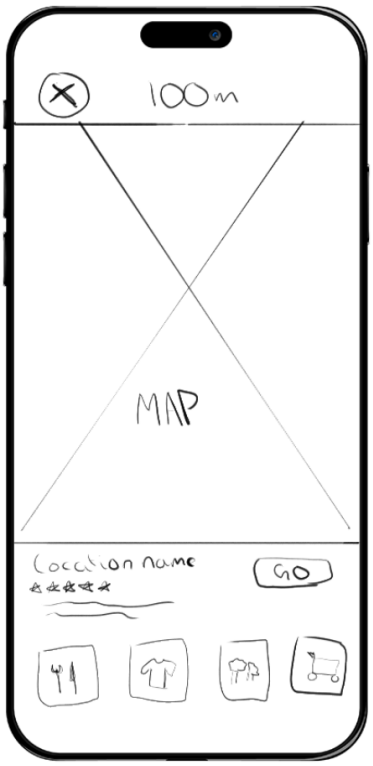


Figure 10 Low-Fidelity Community

As for the community page, the idea was to have a page where users can share their experience of travelling with others.
Making sure the top header and navigation are consistent will allow to build a sense of familiarity. This will make the usability of the app more comfortable (Szerovay, 2022). Again, following the same visual hierarchy as on the home screen.

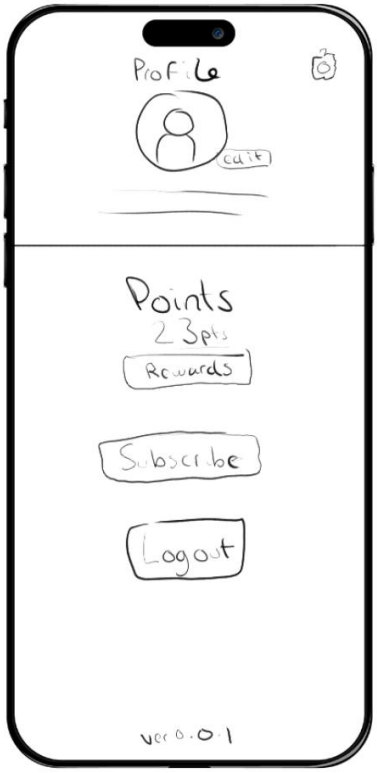
Maps Page:



The maps page was a simple design. Having Jacob's Law in mind, I wanted to implement a mapping page like Google Maps, since millions of people use this app every day. This will allow the users of this app to easily use it, since it will be the same experience as what they are used to (Yablonski, n.d.).

Figure 11 Low-Fidelity Maps

Profile Page:



This page will be a means to have the user check up on their rewards points, join or check their subscription and logout.
Using hick's law, I didn't want to overwhelm the user, so keeping the button options simple will allow them to quickly decide on what they want (Nikolov, 2017).

Figure 12 Low-Fidelity Profile

Subscription Page:

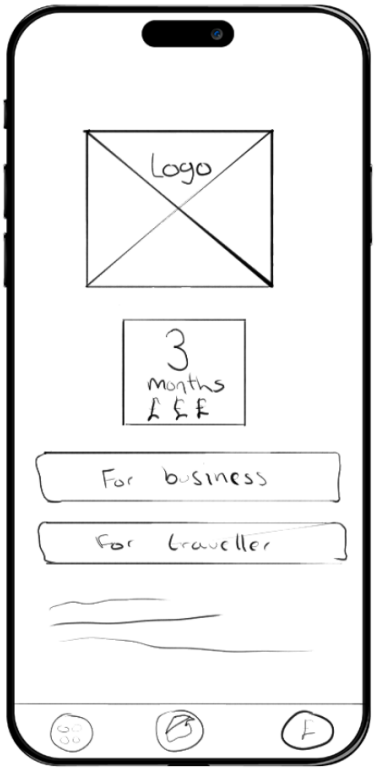


Figure 13 Low-Fidelity Subscription

This will be the page for the user to check the subscription details for the different plans and their prices.
Having the “For Business” and “For Traveller” buttons the same, follows the Law of Similarity, which in this case this will indicate to the user that each button will do the same but for its own plan (Harley, 2020).

Payment Page:

Figure 14 Low-Fidelity Payment

The payment page, I wanted to keep it simple, since so far, every page has been kept simplistic from a visual standpoint. This follows will the consistency principle and the law of similarity, since having all the buttons and text boxes the same as it was on the other pages will allow the user to recognise what to do (Chrbutler.com, 2024).

2.6 Digital wireframes

Login Page:

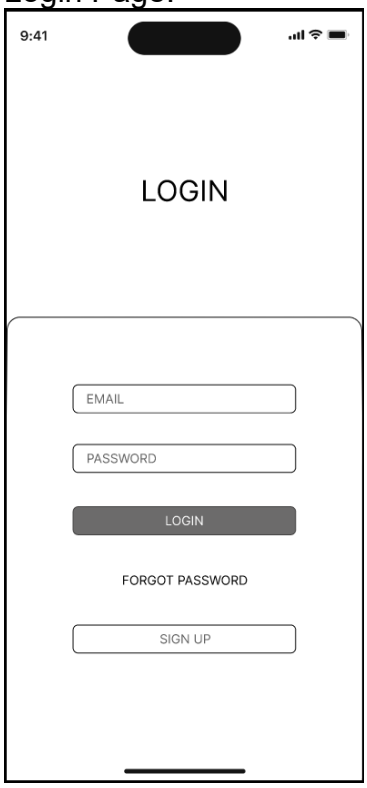


Figure 15 Mid-Fidelity Login

All the digital wireframes were done within Figma.

Using what I already have from my low-fidelity, the following digital wireframes were simple to do.
Within the login screen kept the same for the most part. However, I wanted to differentiate the “login”, “forgot password” and “sign up” buttons. Having different contrast such as the login button, will have the user indirectly focus on this button, since it stands out the most. Not having a box around the forgot button, is another way of using contrast, since this will only stand out when the user is looking for it (CourseUX.com, 2015).

Home Page:

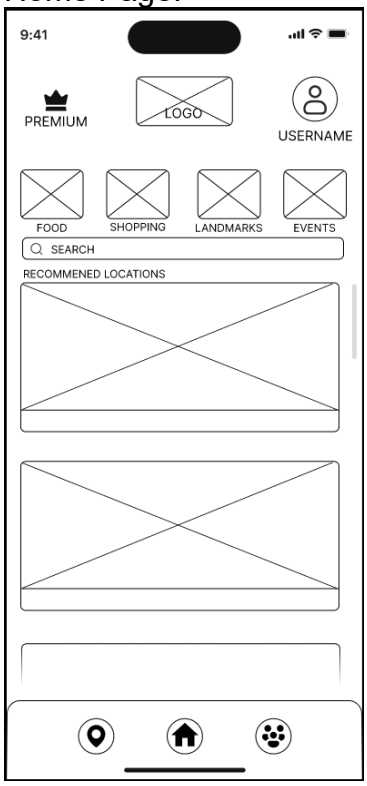


Figure 16 Mid-Fidelity Home

For the home page, there are not many changes, since I was happy with what I had.
As mentioned in section 2.5, the visual hierarchy pattern is the same. Since usually the user will start from the top left side of the screen, I moved the premium icon there, this was just a thought from a business design viewpoint, since I would want as many eyes on this as possible.
Using Jabob’s law, the icons in the navigation bar, should be universally known as the maps, home and community/ a place to share. This will increase the user friendliness of the app, since familiarity leads to ease of use (Yablonski, 2024).

Community Page:

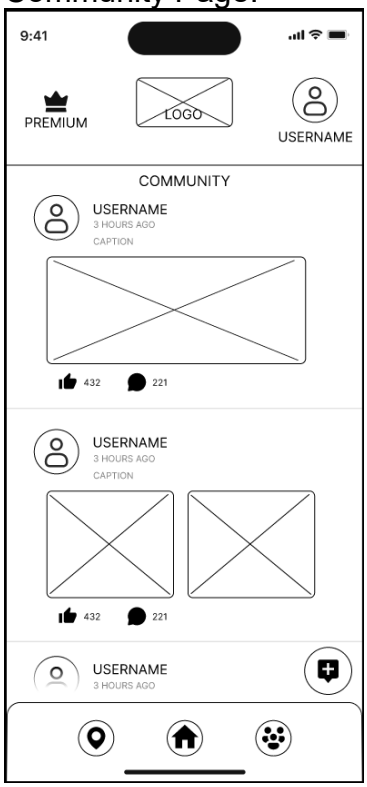


Figure 17 Mid-Fidelity Community

The community page is where many will come to share and interact with others. The top header and navigation bar follow the law of consistency, which is having elements do the same for each page. Having the navigation buttons do the same action, allow the user to learn what they do and create a smooth experience when going through the pages (Hapy Design, 2024).

Maps Page:

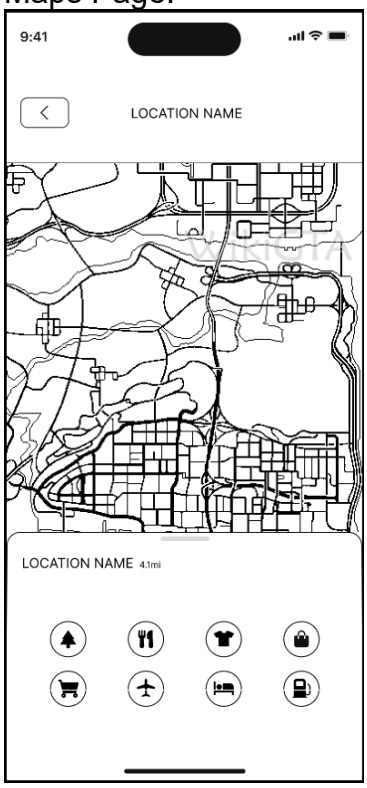


Figure 18 Mid-Fidelity Maps

Having liked the design of the paper prototype, I decided to keep the overall shape of the page the same, just refining some elements to make a more user interactive page. Again, taking inspiration from Google maps.

Profile Page:

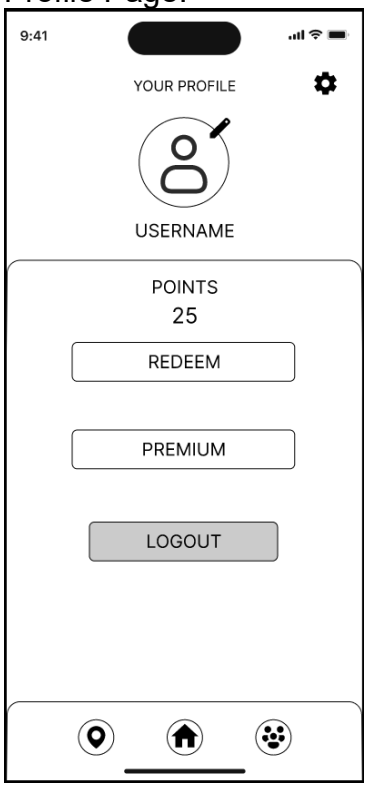


Figure 19 Mid-Fidelity Profile

For the profile page, I removed the top header. This was intentional, since the header only had the app logo, premium icon and user profile picture. Having this in the profile page will be repetitive, due to the page having these elements already.
As mentioned before, within the login page, having the buttons contrast to one another can create clarity. Here, I thought of having the logout button smaller and darker, allowing the user to easily identify and click on it without any mistake (CourseUX.com, 2015).

Subscription Page:

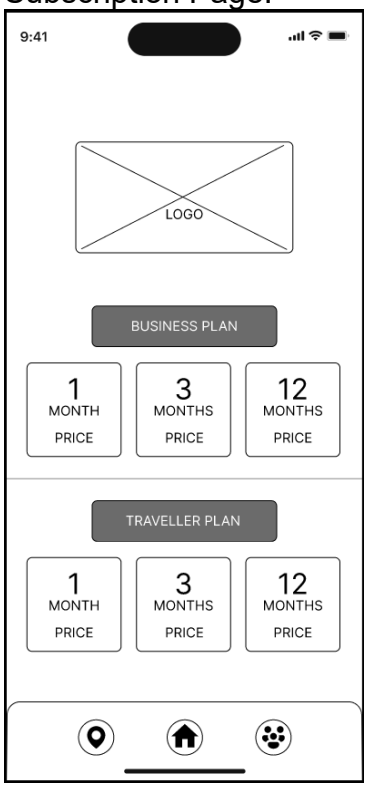


Figure 20 Mid-Fidelity Subscription

The subscription page for the most part was kept the same. For user freedom, I added extra options but made sure not to add too many since Miller's Law states that a person on average can remember around 7 different information at a time (GeeksforGeeks, 2024).

Payment Page:

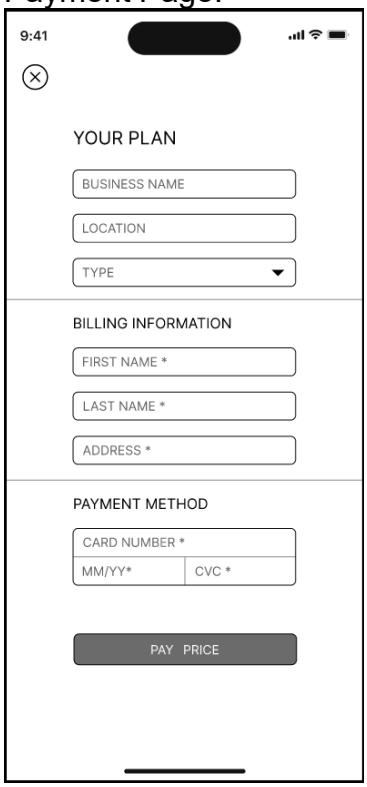


Figure 21 Mid-Fidelity Payment

Taking what I have from the paper prototype, I translated that into the digital wireframe. Not many changes were made other than the clarity of the payment button, the reasoning for this was to highlight the price of the product/ subscription, to prevent user error.
Also segmenting each section with a line and title, as well as, having placeholders will ensure the user to know what needs to be filled out and where, avoiding slips (Laubheimer, 2015).

2.7 High-fidelity prototype (mockup)

The high-fidelity were all done using Figma and the frame size being that of the iPhone 16 Pro Max (440 x 956).
The main colour scheme I chose for my app was orange (#FF9500), blue (#407DF7) and white (#FFFFFF). This was due to one of the personas being deuteranopia colourblindness, thus using a palette of blues and orange would help differentiate the elements on the pages and be easy on the eyes (Shaffer, n.d.).

Login and Sign-Up Page:

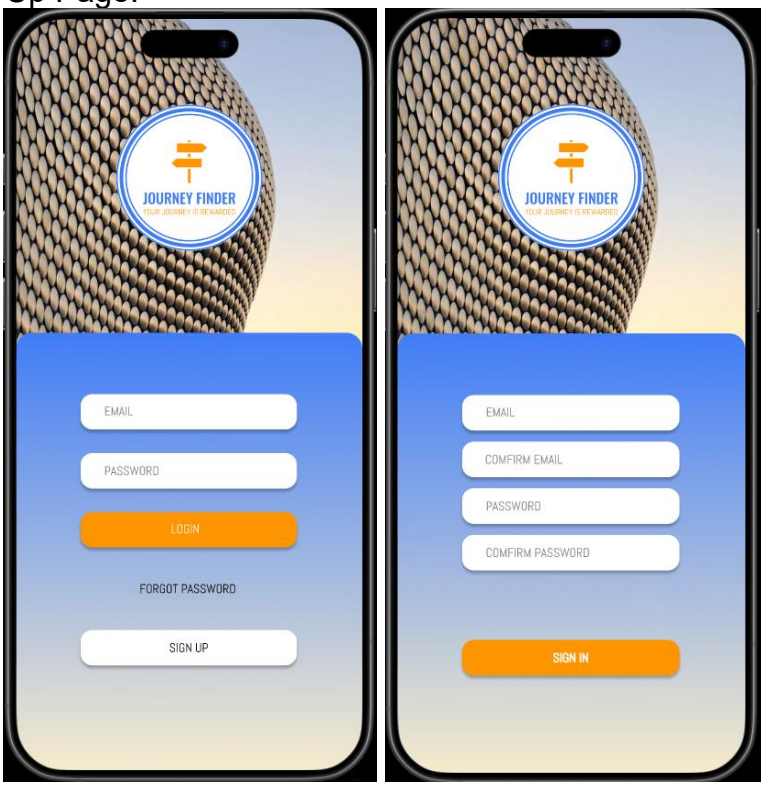


Figure 22 High-Fidelity Login and Sign-Up

Using the digital wireframe, I ended up with this design for the login screen, also translating it into a sign-up page. For both pages I wanted the colour orange to be the indicator for moving forward, since it has the most contrast to the blue background. Doing it this way will enhance the importance and capture the attention of the user to the button, whereas the “forgot password” button is almost hidden compared to everything else (Incharaprasad, 2023).

Home Page:

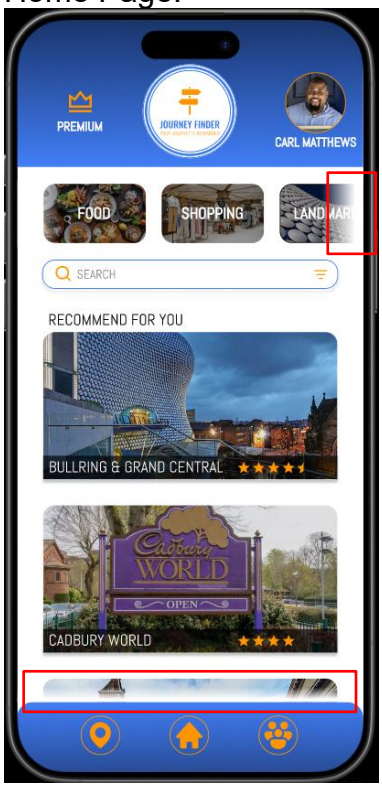


Figure 23 High-Fidelity Home

Following the same colour scheme and using white as the background allows the other elements to stand out. As for the top tabs and the recommended section, I thought of adding a slight fade to ends (highlighted by the red box), to give the user a visual que that they can horizontally and vertically scroll for more options. Using a feature like this will reduce the cognitive load on the user, since it avoids clutter on one page and sticking to the rule of 7 stated in Miller's Law (Whitenton, 2013).

Community Page:

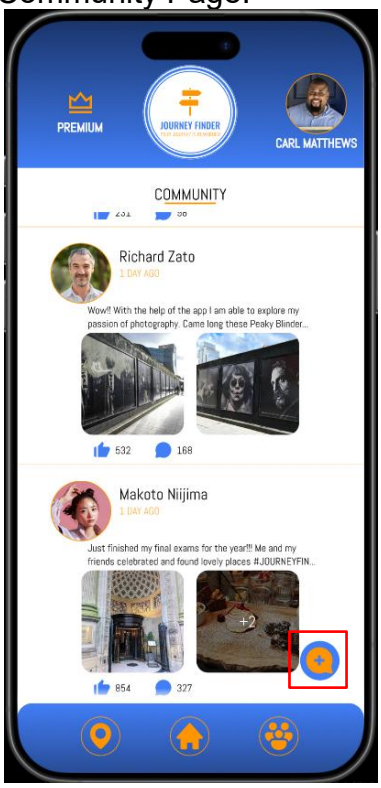


Figure 24 High-Fidelity Community

This community page design was heavily influenced by X, formerly known as Twitter. (Duarte, 2024) states X has over 600 million active users monthly and 58% of these users are under 35 years old.
The reason for this was due to Jacobs Law, which states that users use other sites more, so having my community page work like X, users will already know how to use it (Nielsen, 2019). Since one of my personas is a 21 student and another being a photographer, having a page like this would be appropriate.

Figure 25 X (Twitter) Add post feature

Maps Page:

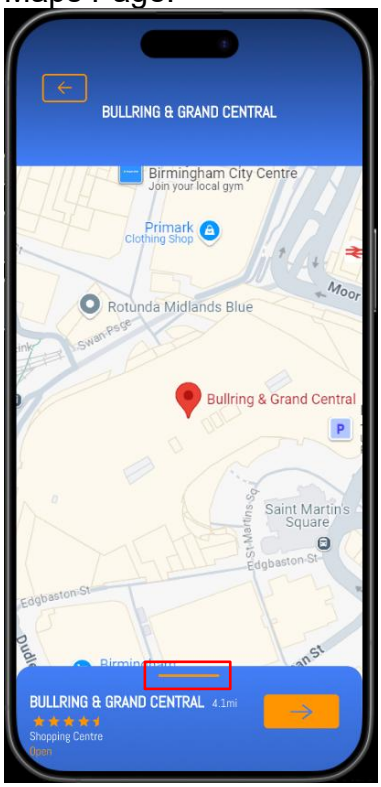


Figure 28 High-Fidelity Maps

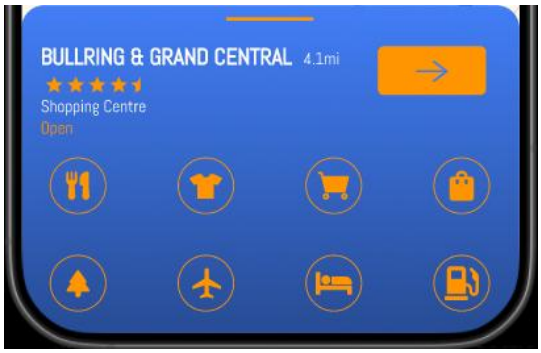


Figure 27 Maps Extension

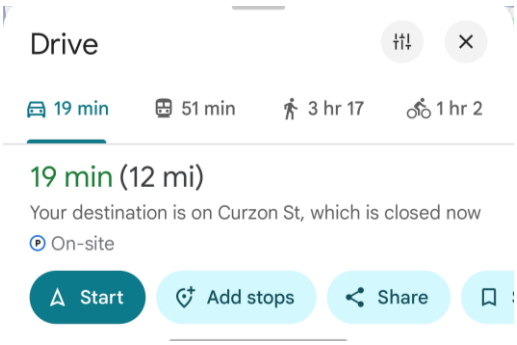


Figure 26 Google Maps

Similar to my thought process for the community page, here I used the mobile app version of google maps as a reference. Since over a billion people use Google Maps monthly (Russell, 2019), designing my page after it would allow for an easy learning curve and familiarity, for example the red box highlights a line, which many users will recognise and have them pull up an extension. Comparing figure 26 and 27, shows the inspiration of where I got the design, but it had differences such as mine has tabs that is intended to filter to what the user is looking for.

Profile Page:

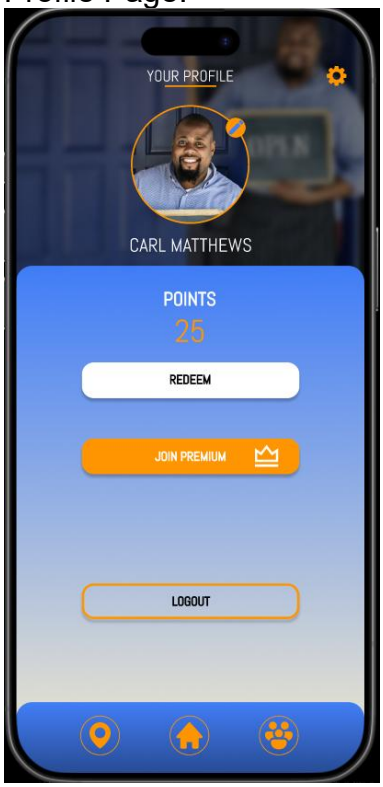


Figure 30 High-Fidelity Profile

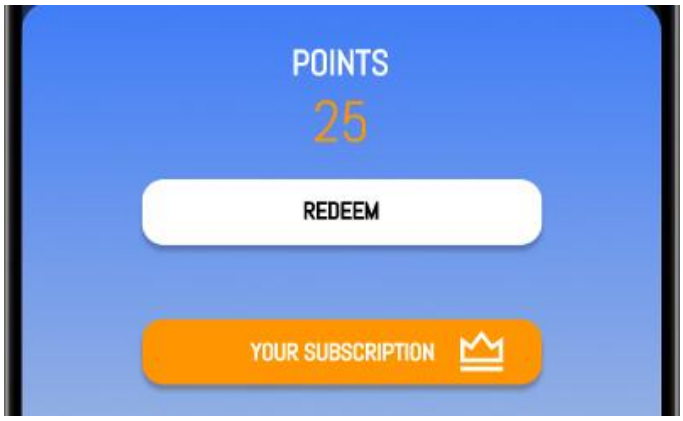


Figure 29 Profile after Premium

Other than using universal icons such as the pencil to edit the profile picture or a cog for the settings. To create a visual hierarchy with colour, I decided to have the "Redeem" button white due to it grabbing the user's attention the most. Due to the points feature mentioned in the user scenarios, this will make it easier to see how many points they have and what button to press. Having the logout button transparent and separated from the other two will reduce errors like accidental logouts (Gordon, 2021).

Subscription Page:

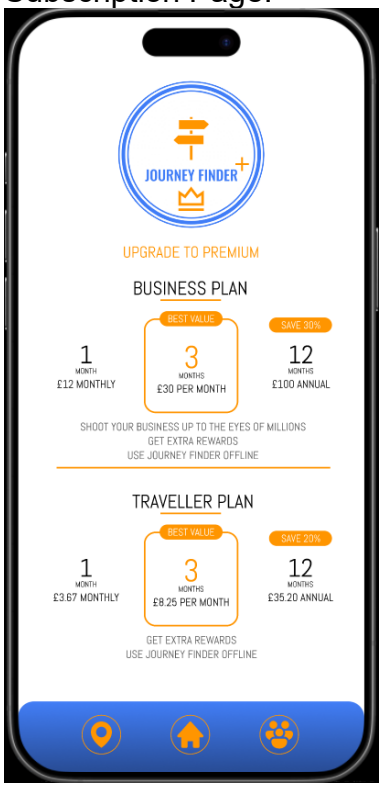


Figure 32 High-Fidelity Subscription

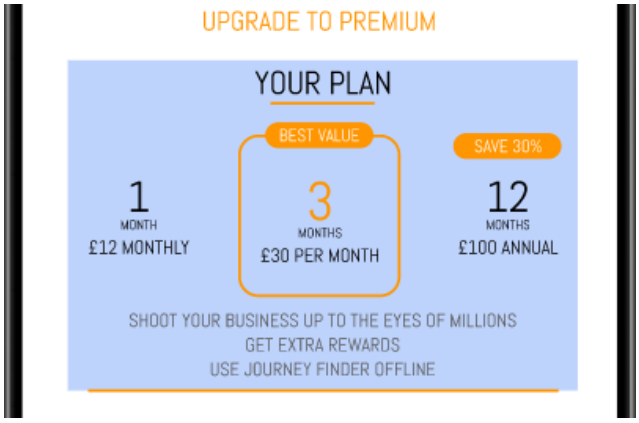


Figure 31 Subscription after paying

Staying consistent with the colours, here I again used orange to capture the user's attention highlighting what is of best value. Also making sure the text for the subscription is straight to the point. This was a thought for all text within the app, making sure it's readable and aesthetically pleasing to the user (The Interaction Design Foundation, 2024). Though the design looks minimal and simplistic it still conveys a lot of information without increasing the cognitive load on the user.

Payment Page:

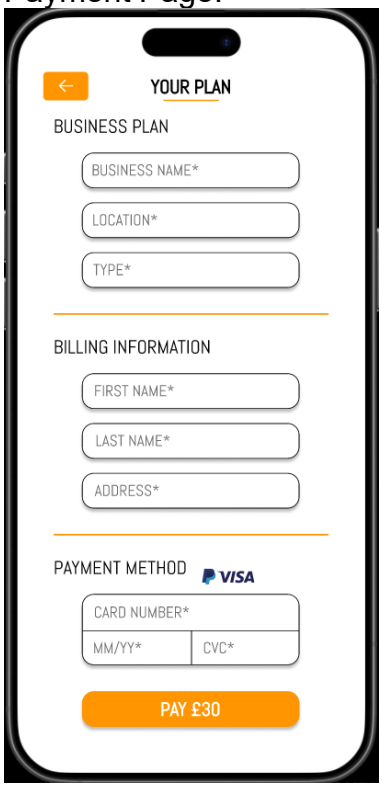


Figure 33 High-Fidelity Payment

Labelling and separating the different sections will reduce the cognitive load on the user, and allow them to fill this section without complications, this is known as chunking (Yablonski, 2024).
The button on the top left will be recognised by many as the “go back” button, this is out of the way but easily accessible since its not too small. Fitts’s Law is also implemented in the bottom navigation within the other pages, they are easy to use with one hand and spaced out to prevent accidental presses (IDF Instructor, 2019).

3 Usability testing

Usability testing is a way to assess how your product will work in the real-world, by having actual users, not designers or developers, test it. This will help find mistakes or give information on what to improve (Maze, n.d.).

3.1 Protocol

For the usability testing all the tests were conducted using maze. I had family ask friends that I may not have known to avoid bias. For two of the participants, it was done over an online chat using WhatsApp, the other 3 were done in person. Each participant were asked to create an account, logout of the app, check the community page, become a member, use the maps page and explore the whole app.
The tests were done using iPad, phones and laptops. I just took notes using my phone.

3.2 Data collection methods

All quantitative data will be collected using Maze. This Quantitative usability testing is perfect since it will allow me to gather information on how long, the success rate, the analytics of where the user clicked (heat maps) and the tasks they struggled on (Moran, 2018 and Maze, n.d.).
As for qualitative this was also done on maze, but it was a single question at the end about how they got on. I also followed it up with 5 basic questions about their thoughts, this will be discussed within section 4. Due to timing issues, only 3 of the participants were observed in-person, this allowed me to see how the user got on and this will also encourage them to test the app properly, due to me being there (Springboard Blog, 2021).

3.3 Participants

When looking for participants I didn't want to just have family and friends, so I asked my family if any of their friends were able to do the test, I managed to get 2 participants who I did not know, the others were family and provided them with a consent forms. I know it can lead to bias results, so I asked them to be as realistic as possible. The participants are also from a spread of different ages, so I can observe their interactions with the app.

Participant No.	Name	Age	Occupation
1	Asma	50	Carer
2	Aadil	18	Student
3	Farihah	23	Student
4	Rasheda	28	Paralegal
5	Bushra	25	Primary Teacher

3.4 Results

Before the tests were live, I did a quick pilot test to see how everything would go. This was done on desktop, so there were no problems. However, the first two participants that did the test, was done using a mobile device, this bought inconveniences, since when the user was tapping around seeing how the app works, whenever they tap near the left side or the right side of the screen it would take them to another page. This was evident during my observations, which I can see the participant was somewhat frustrated (during becoming a member task), to where they dropped off and didn't complete the task. Phone usage problems shown in figure 35 to 37 in the appendices. Due to not knowing the reason for this, the other 3 tests were done using a laptop.
Overall, the data from Maze shows a rating of 50% (figure 38), however this doesn't fully represent the actual usability.

Task 1 was to create an account, this came back with a 100% success rate with 12 seconds taken on average and a 66.7% misclick rate (figure 39). The score given was 42%. However, most of these misclicks were out of the tester's curiosity. Observing them showed me they will click on elements that do nothing to see if it will do something, for example clicking on the email and password text box to fill it out (figure 40).
Task 2, shown in figure 40, that the logging out task gave the lowest misclicks with 100% success. The fastest completion being that of 4 seconds (figure 42).
Task 3 again with a 100% success rate, but showing a 53% misclick, these misclicks, same as task 1, came from users clicking on non-interactable elements (figure 43 to 44).
Task 4, this is where I previously discussed the problems I came across with testing using a phone. 80% success, 20% drop-off and 58% misclick (figure 45, 46).
Task 5 (Use maps) and 6 (Explore the app) I set as "Free Explore", the reason for is I wanted to observe how they would go about using the maps page and if they'd be able to use with ease due to its design being similar to Google maps. Figure 47 to 48 will show the heat maps. Task 6 was to see how familiar they got with the app after doing the tasks.

As for Qualitative, at the end of the tests I left a question of their overall thoughts on the app prototype and mentioned before, observing them in person seeing how they react verbally and non-verbally (figure 49 to 54).

4 Analysis/Discussion

To conclude this report, after completing the usability testing, the feedback for the most part was positive, showing that the participants had ease when using the app how it was meant to be used. The testing also brought to light problems that affected the comfortability and enjoyability of the product, which I discussed within the results. Figure 36 shows a heat map of a user's tab, looking upon this revealed to me that even though they tapped between the icon and the text it still didn't take them to the page it was intended to, meaning that I should work on and improve the elements such as this, so the area of effect is taking up a larger area around the icon and not just icon itself.

In addition to this the feedback shows I have designed the app in mind of 'ease of navigation', as users said it was "easy to navigate" and "clear indication of where each segment can be found". However, one said the icons can titled to give them confirmation of where it will take them. And another major feedback was adding a "Dark Mode feature", which is something I completely overlooked, since I was so focused on using colours that are friendly to those with colourblindness. To show what dark mode could look like I quickly designed a concept idea of a "dark mode home page". Changing the white background to a dark blue (#081D47) and darkening the top header and bottom navigation bar, really does feel easy on the eyes as well as looking pleasing.

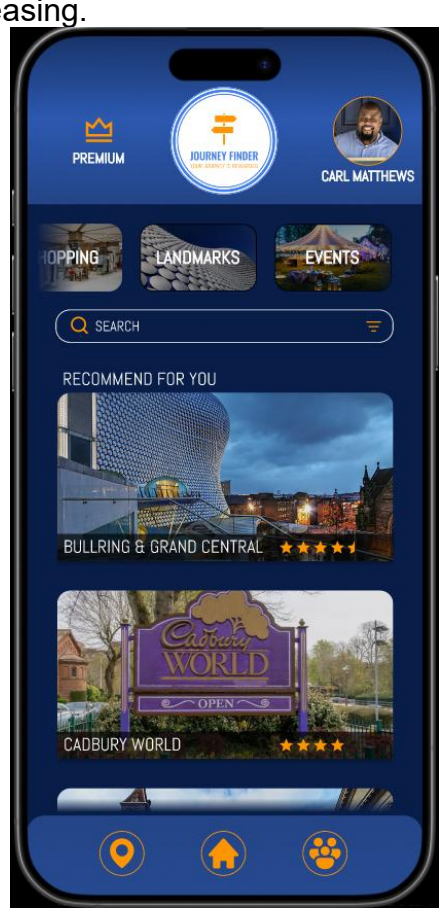


Figure 34 Implementing Dark Mode (Concept)

Other Usability testing methods I could use are A/B Testing, where creating two versions of the of the app and sending them out to test live. Such as, using the feedback I got, I could have had some participants test the dark mode with titled icons and the other group of participants with the original. Once getting the results, compare which performed better. This type of testing is a strong way to know what works and what doesn’t at a low cost, and it isn’t too difficult to gather data from (Moran, 2018). However, it can take a lot of recourses to get started since you will be developing two different interfaces (Rees, 2010).

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Appendices

Link to Figma Prototype:
https://www.figma.com/proto/ucBppxYKnkH4W2RkWEXQIr/Hi-Fi-App?node-id=2-2&t=Az52t8uucfePpICn-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1

Testing Using Phone Errors:

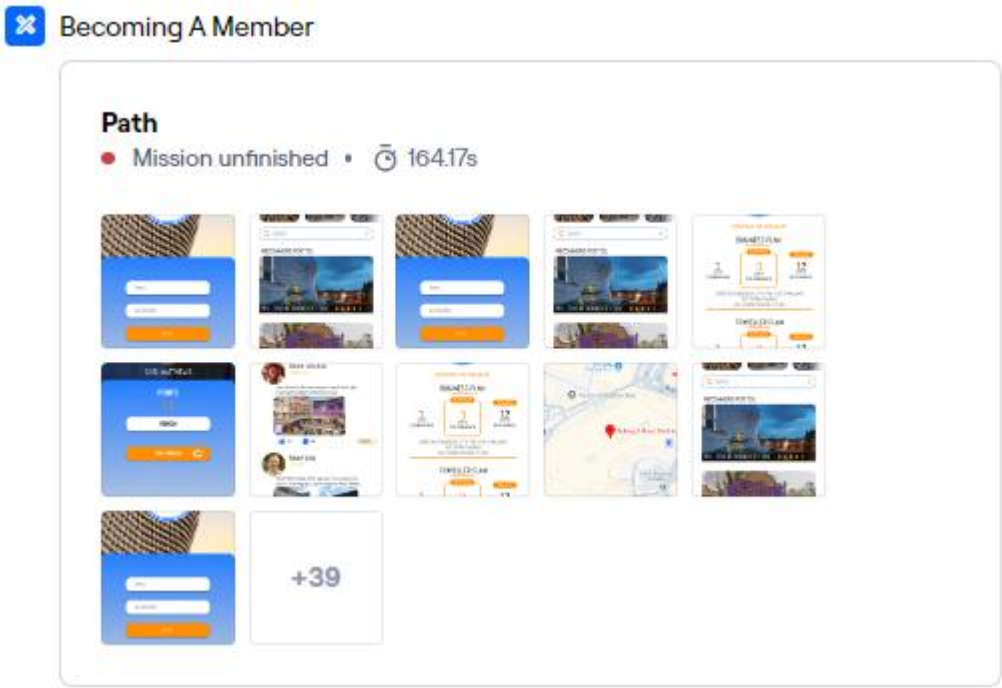

Figure 35 Participant 1 Incomplete Task

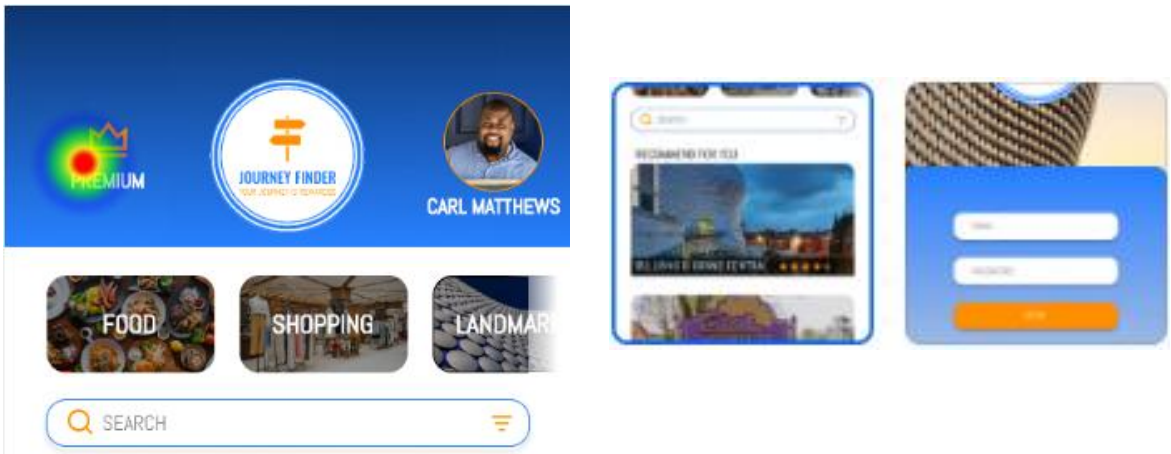

Figure 36 Heat map shows slight misclick takes them back to login screen

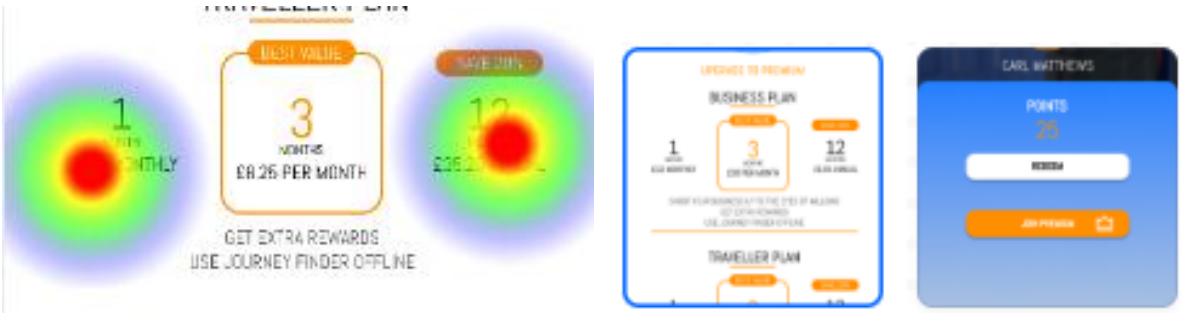

Figure 37 tapping on non-elements that are on the left or right causes them to go to another page

Quantitative Data:

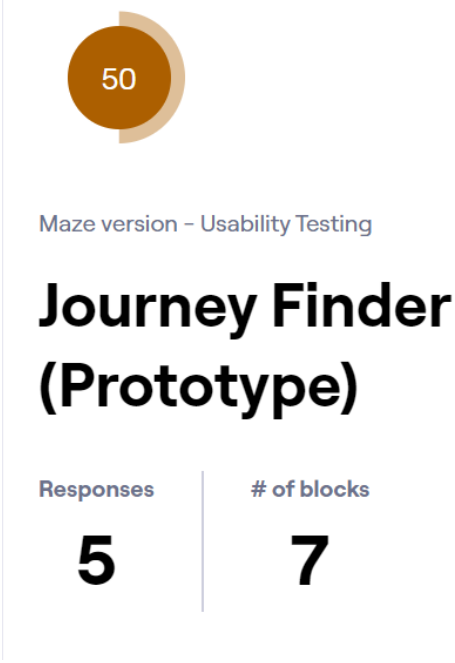

Figure 38 Maze Usability Score

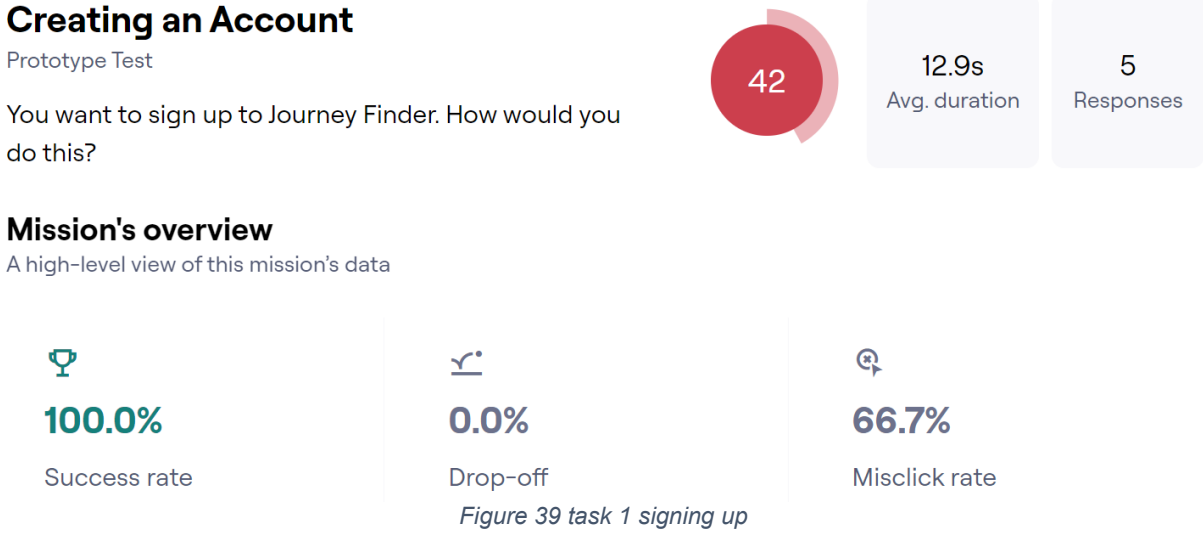

Figure 39 task 1 signing up

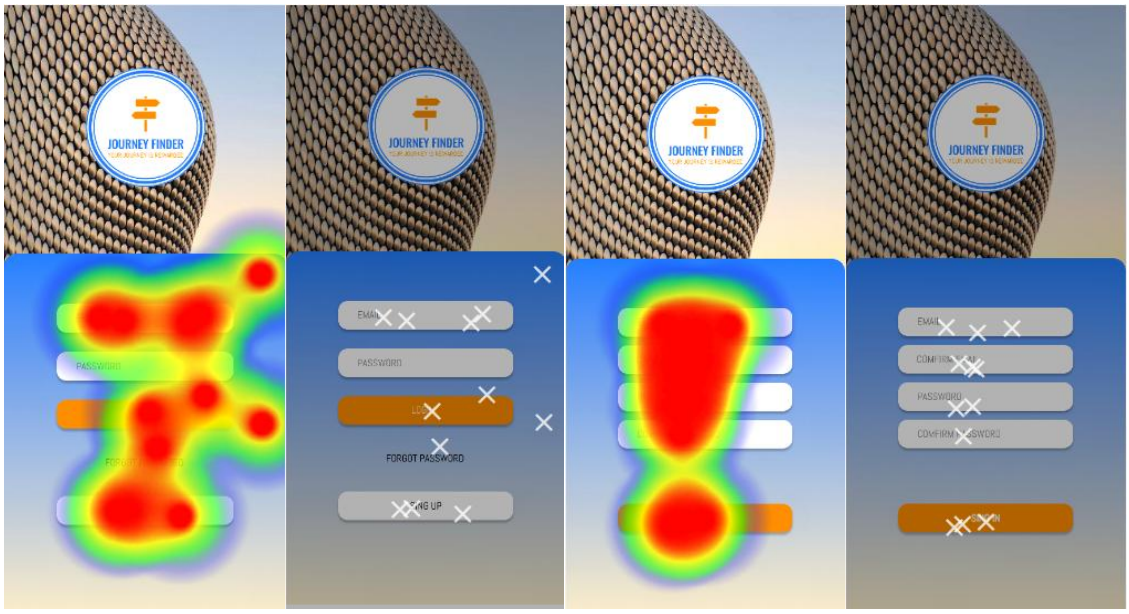

Figure 40 Login/Sign up page "misclicks"

Logging Out

Prototype Test • Screen-based

Once finished using the app, you want to log out. How would you do this?

15.6s	5
Avg. duration	Responses

100.0%	0.0%	15.4%
Success rate	Drop-off	Misclick rate

Figure 41 Task 2 Logging out

Path

Success • 3.18s

Figure 42 Fastest Logout

Check Out the Community

Prototype Test • Screen-based

You want to share and interact with the community, using Journey Finder. What page would you do this on?

22.9s	5
Avg. duration	Responses

100.0%	0.0%	52.9%
Success rate	Drop-off	Misclick rate

Figure 43 Task 3 Community page

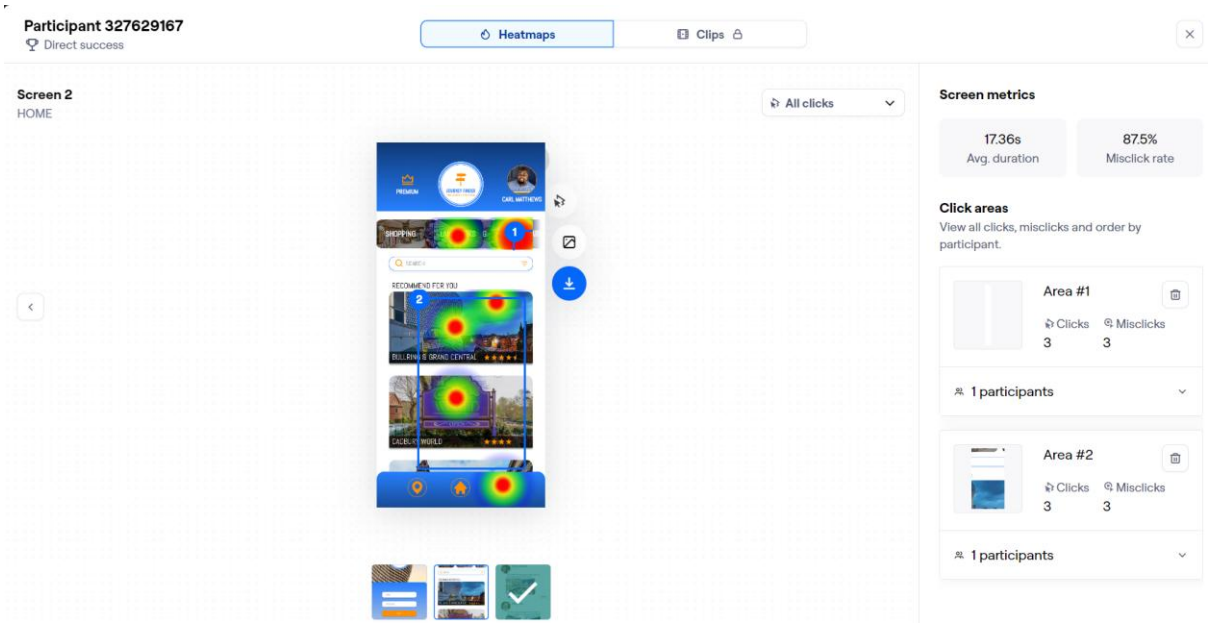

Figure 44 Testing (exploring) leads to misclicks

Becoming A Member

Prototype Test

As of now, you are a non-member, you've seen the preimum plans and want to sign up for it. How would you do this?

58

27.3s	5
Avg. duration	Responses

Mission's overview

A high-level view of this mission's data

80.0%	20.0%	58.1%
Success rate	Drop-off	Misclick rate

Figure 45 Task 4 become a member

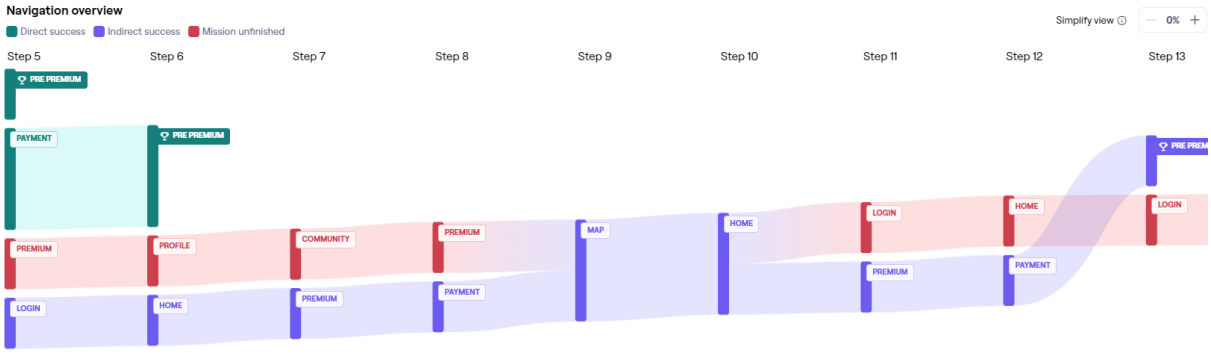

Figure 46 problem leading to drop-off

Map Exploration

Prototype Test · Free Explore

You've found a location to visit and want to use the in-app map. How will you find the map? (Explore the map page. Once done end the task)

68.1s
Avg. duration

5
Responses

Explore

Prototype Test · Free Explore

Look through the various pages. This is relevent for the last question. Once done end task

43.1s
Avg. duration

5
Responses

Figure 47 Task 5 and 6 explorations

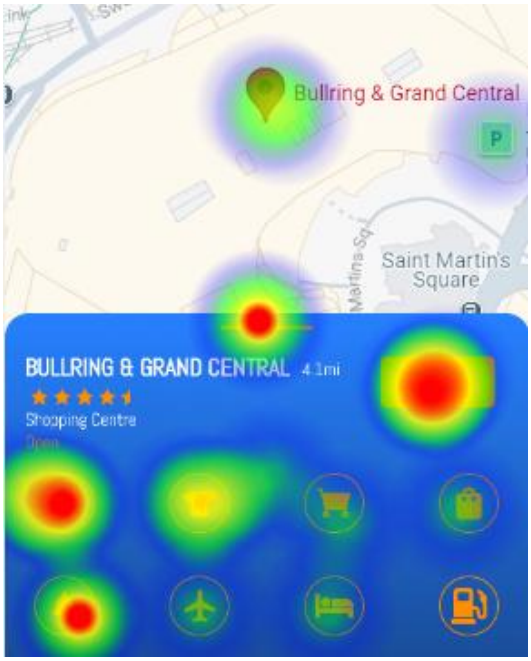

Figure 48 Using the map as intended